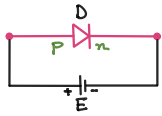


## Tutorial 2

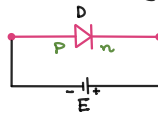
**Biasing:** Applying DC voltage across a device.

**Diode Biasing:** As a Diode has two terminals, two types of Biasing is possible.

① Forward Biasing:



② Reverse Biasing:



We will learn about the biasing principle in detail in EEE309.

[https://youtu.be/\\_Fw--tf-EG4?si=wfCoS4oQGFOW1udB](https://youtu.be/_Fw--tf-EG4?si=wfCoS4oQGFOW1udB) (If you still want to know)

For now, we need to know,

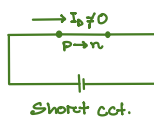
**For Forward Bias:**  $I_D = I_{Diff.} - I_{Drift}$  ;  $I_{Diff.} \gg I_{Drift}$   $\therefore$  There will be flow of current.  
 $\xrightarrow{p \rightarrow n}$  (mA)  $\xleftarrow{n \rightarrow p}$  (nA-pA)

**For Reverse Bias:**  $I_D = I_{Diff.} - I_{Drift}$   $\therefore I_D = -I_{Drift}$   $\therefore$  There will be negligible flow of current ( $\because I_{Diff.} \gg I_{Drift}$ ).  
 $\xrightarrow{p \rightarrow n}$  (mA)  $\xleftarrow{n \rightarrow p}$  (nA-pA)

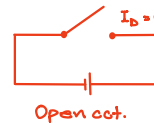
To summarize:

|      | Current? |                                                                |
|------|----------|----------------------------------------------------------------|
| F.B. | Yes      | } So, we can refer to the switching action of diode from here. |
| R.B. | No       |                                                                |

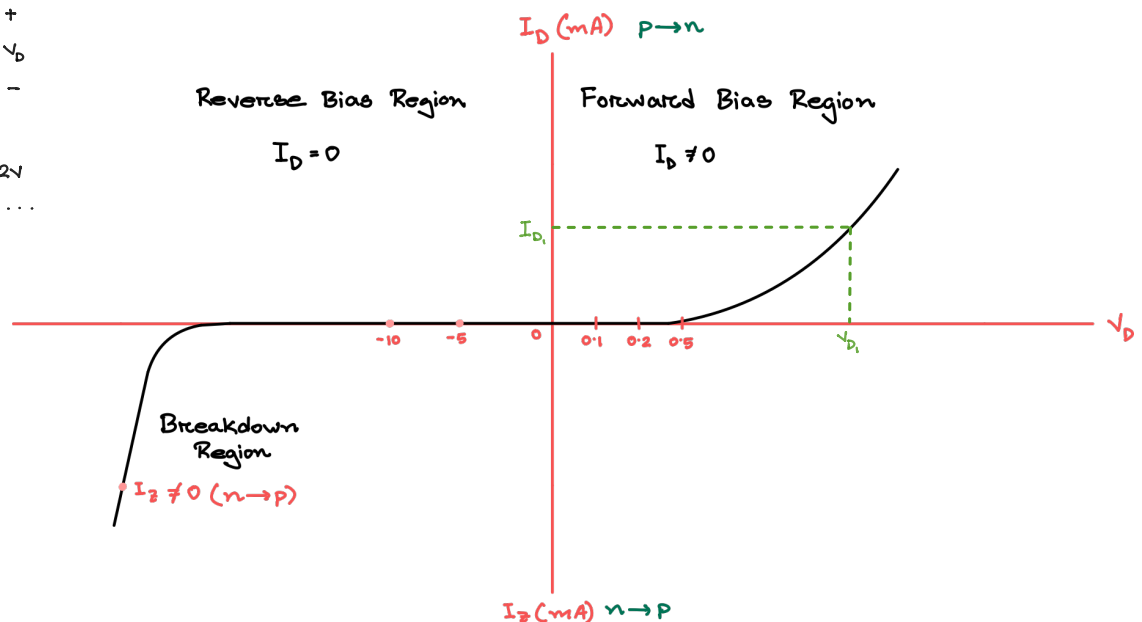
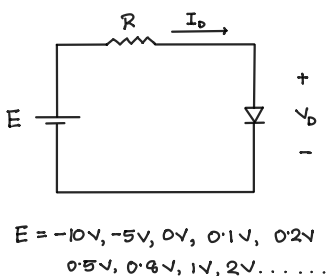
At F.B:



At R.B:



### I-V Characteristics of a Diode:



## Diode Equation:

$$I_D = I_S \left[ e^{\frac{V_D}{nV_T}} - 1 \right]$$

where,

$I_S$  = Reverse Saturation Current ( $I_{\text{briff}}$ )

$V_T$  = Thermal Voltage =  $\frac{KT}{q} \approx 25 \text{ mV}$  (At  $T = 300 \text{ K}$ ) ;  $K$  = Boltzmann Constant

$n$  = Ideality Factor : Si  $\rightarrow 1 \sim 2$ .

$q$  = Charge of Electron

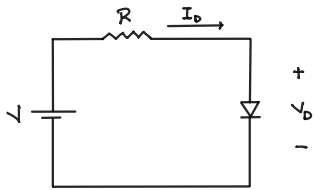
Now,

$$I_D = I_S e^{\frac{V_D}{nV_T}} - I_S = I_{\text{diff}} - I_{\text{briff}}$$

$$\therefore I_D = I_S e^{\frac{V_D}{nV_T}} \quad [\because I_{\text{diff}} \gg I_{\text{briff}}]$$

## Diode Model:

### Piecewise Linear Model:



Here,  $r_D$  = Internal Resistance

Now,  $OR = OP + PR$

$$\Rightarrow V_D = V_{D_0} + PR \quad \text{----- (i)}$$

Again,

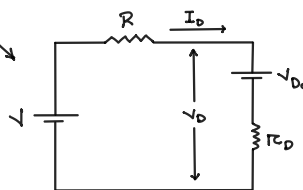
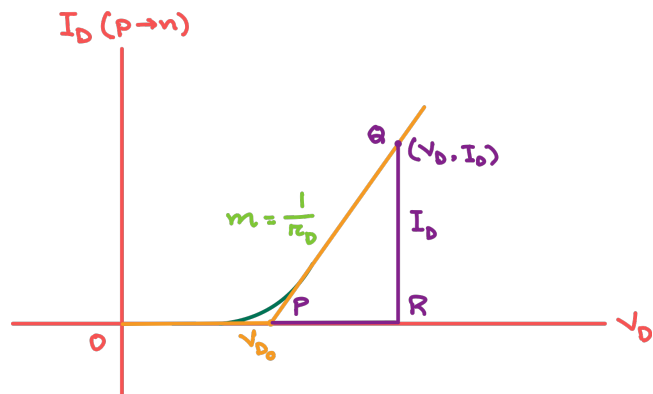
$$\text{Slope, } m = \frac{1}{r_D} = \frac{OR}{PR}$$

$$\Rightarrow \frac{1}{r_D} = \frac{I_D}{PR}$$

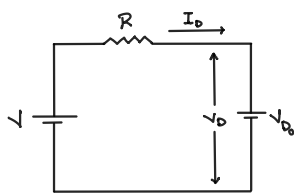
$$\therefore PR = I_D r_D$$

Substituting  $PR = I_D r_D$  in (i),

$$\therefore V_D = V_{D_0} + I_D r_D$$



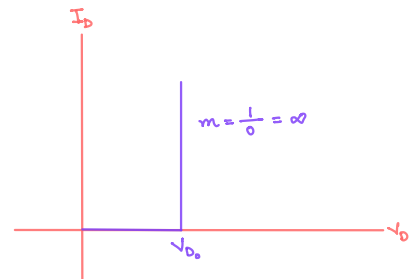
### Constant Voltage Drop Model:



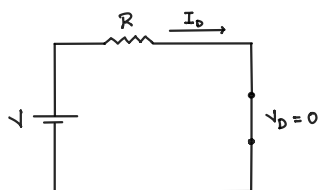
Will use in maths.

Ignoring  $r_D / r_D = 0$ .

$$V_D = V_{D_0} + I_D r_D \quad \therefore V_D = V_{D_0}$$



### Ideal Diode Model:



For switching.

Short ckt  $\rightarrow$  F.B.

Open ckt  $\rightarrow$  R.B.